Herbert Federer

GEOMETRIC MEASURE THEORY

List of corrections

May 1987

page	line	correction
11	8	$\mathrm{Hom}(V,\mathbf{R})$ (boldface $\mathbf{R}$)
15	17	$\underline{p+q}$ (italics)
20	11	in case
20	32	$\mathbf{1}_{\Lambda_j V}$
22	9	for $\varphi \in \Lambda_t V$
22	11	$\Lambda^1 V$
26	24	$\mathbf{i}$ (boldface)
34	29	$\langle \xi, *\beta \rangle = (-1)^{p(n-p)} \langle *\xi, \beta \rangle$
34	30	replace period by " $,\beta \in \Lambda^{n-p} V$. "
35	3	$(v \otimes w) \bullet (v' \otimes w')$
57	26	$\mathrm{card}(A)$
59	3	$a \in \alpha$
60	4	$\underline{\text{In case}}$ $\varnothing \in S$ $\underline{\text{and}}$
63	2	finitely many open sets
63	30	only if there exists
65	20	we mean
66	13	$S_1, S_2, S_3, \ldots$
66	15	$\mathcal{N}_j$ (script)
67	2	$\cup G$ $\underline{\text{and}}$ $\cup H$
68	14	subset of $Y \times \mathcal{N}$
76	3	$\underline{\text{and}}$ (italics)
78	7	consider the set
86	26&27	On the space $\mathbf{A}(\varphi, Y)$ [see 2.3.8] we define the functions $\varphi_{(p)}$ for $1 \leq p \leq \infty$ by the formulae
86	insert at the bottom the following two lines:	
		The functions $\varphi_{(p)}$ are seminorms on the vectorspace $\tilde{A}(\varphi, Y) = A(\varphi, Y) \cap \{ f : \mathrm{dmn}\ f = X \}$.

page	line	correction
87	1	This is obvious in case $p = 1$ or $p = \infty$;
87	18	subset of $\mathbf{L}_p(\varphi,Y)$, provided $p < \infty$ or $\dim Y < \infty$.
87	32	$\varphi_{(1)}$ closed subspace
88	3	because the subspace
89	19	$u \in \mathbf{L}_q(\varphi)$ and $\varphi_{(q)}(u) \leq 1$
91	10	$\mathbf{L}_1(\varphi,Y)$
91	11	Proof. In case ...
91	28	if f and g belong to L, so does ...
98	10	μ: $\mathbf{L} = \tilde{\mathbf{L}}_p(\varphi) = \mathbf{L}_p(\varphi) \cap \{ f : \operatorname{dmn} f = X \} \to \mathbf{R}$
98	11	$\varphi_{(p)}(f) \leq 1$
98	14	measurable;
99	13,14,17	$\mathbf{L}^+$
100	3	Daniell integral on $\tilde{\mathbf{L}}_\infty(\varphi)$ such that $\mu(f)=0$ whenever $f \in \tilde{\mathbf{L}}_\infty(\varphi)$ with $\varphi_{(\infty)}(f)=0$, then there exists
100	5	$\tilde{\mathbf{L}}_\infty(\varphi)$
100	7	$f_n \in \tilde{\mathbf{L}}_\infty(\varphi)$ with $\lvert f_n \rvert \leq 1$ and
100	9	$g \in \tilde{\mathbf{L}}_\infty(\varphi)$ with $0 \leq g \leq 1$ and
100	21,22,23,24	$\tilde{\mathbf{L}}_1(\varphi)$
100	22	$\mathbf{L}^+$ is dense in $\tilde{\mathbf{L}}_1(\varphi)^+$
101	15	$x \in \mathbf{R}^2$
102	8	$\tilde{\mathbf{L}}_1(\alpha_1 \times \beta_2) \cap \tilde{\mathbf{L}}_1(\beta_1 \times \alpha_2)$
102	12	$\tilde{\mathbf{L}}_1(\varphi)$
107	24	$\tilde{\varphi}(v)$
118	20,21,24	Replace " $\mathbf{L}_1$ " by " $\tilde{\mathbf{L}}_1$ "
119	3,4,6	
119	19	measures, then $\nu(h) = \int h \, d(\alpha \times \beta)$ for $h \in \mathcal{K}(X \times Y)$.
124	30	Whenever $0 \neq v \in \mathcal{K}(X)^+$, $(\operatorname{spt} v) \times (\operatorname{spt} v) \subset \beta$, $a \in \operatorname{spt} v$
125	7	insert period at end
125	8	delete "whenever $v \in S(\beta)$." and "Closure"

Page	line	correction
131	10	measure over G associated with a nonzero monotone left (right) χ covariant integral is a
132	22&25	$f \in \Psi$
133	3	$U = (G \sim P) \cup (P \cap U)$
133	4	measure over X associated with a nonzero monotone χ covariant integral is a Radon measure.
134	26	$\mathcal{L}^n$
135	35	$(1-t^2)^{(n-1)/2}$
136	7	f on $\mathbf{R}^n$
136	29	$\|\det\|^{-n}$
138	34	$\|\det[L(g)\|W]\|^{-1}$
141	25	$\chi(g) \to \chi(h)$
142	33	$G_0 = \emptyset$
143	17	function on f, $\emptyset \notin F$, and
145	3	[F 12,
146	22	$\rho(x,c)/\rho(a,c)$
151	24	V(X), φ is positive on V(X), $1 < \tau < \infty$
151	29	$\{(x,S)$:
152	10	2.8.3
155	26	$\mathbf{D}(\Psi, \varphi, V, x) \notin \mathbf{R}$
158	21	V (italic)
159	7	$X \sim \{z: f(z) \leq t\}$ has density 0 at x.
159	19	(φ, V)
161	11	2.8.14
162	24,25	$\mathbf{B}(x,2r)$ (boldface B)
163	11	$A \cap S_j = \emptyset$
170	24,35	[F 12,
174	10	have been constructed
177	23	$\int^*$
177	27	[F 12,
184	8	space of all nonempty compact convex subsets of X.
184	10	compact subsets of X,

page	line	correction
188	21	$\mathcal{J}^m$ (script)
192	4	card(H_0)=1, for $j \geq 1$ each member
193	7	$\sum_{T \supset S \in \Phi(J,n)}$
194	8	$\sigma_j \geq t_{j+1} \to 0$
195	20	, then
197	4	$\mu\alpha(m)^{-1} 2^m$.
206	13	separable, complete metric space X,
210	26	$u_{n+1}-u_n=g(u_{n-1})-g(u_n)$
214	24	$\geq 1/j$.
218	12	$U(z,1/j)$
219	4	$\{|\varphi(v_1,\ldots,v_m)|:$
221	1	$a+t(x-a)$
224	4	$\alpha h(x)+(\beta+1)h(s)$
225	11	supports form a
226	20	$\sum_{j=0}^{i}$
228	19	all a in A
230	15	It follows
232	11	and maps $f_i: W_i \to B$ of class k such that $f_i(x)=x$ for $x \in B \cap W_i$.
232	14	$\to T \cap B$
232	15	is a map of class k for which $\rho(x)=x$ whenever $x \in B \cap T \cap f_{i-1}^{-1}(T)$,
234	12	with (Roman)
235	19	such that $f(a) = \eta$ in case $a \in S$ and
242	23	$\Omega(n,m)$
244	11	$J_m f(x)$
246	3	$f(a) \leq f(b)$
246	13	is $\mathcal{L}^1$ summable
248	21	$\mathcal{L}^{m-n}$
248	25	$\mathcal{H}^{m-n}$

page	line	correction
249	21	$\mathcal{L}^m(A)$
253	29	$\delta_{i+1}<\delta_i/2$;
253	32	relative to T at a; in case $\Phi\{a\}>0$ this holds with T replaced by $T\cup\{a\}$.
254	2	replace "coincides with" by "generalizes"
257	26	$\mathcal{L}^\mu(B)$
258	20	add " $R=W\sim\bigcup_{i=1}^{\infty}\Psi_i(K_i)$, "
258	22	$P_{i,j}\cup R$.
259	3	Furthermore $\mathcal{H}^m(R)=0$, hence 2.10.25 implies
259	4	$\mathcal{H}^{m-\mu}[R\cap f^{-1}\{z\}]=0$ for $\mathcal{H}^\mu$ almost all z in Z.
261	4	$\mathcal{H}^m$
261	22	in 2.10.5, choose
261	29	x-w
262	14	help of 3.2.20 that
265	3	delete "because $\mathbf{O}(n)$ is compact,"
265	12	$\mathbf{O}(m,m-1)$
270	25	hence by 3.1.24 the set
270	28	and 3.1.18 shows that
271	24	$g(x)^{-1}x$
273	18	m-th
275	2	r^{n-j}
275	14	$\{z:\ \mathbf{dist}[z,f(C)]<r\}$
279	27	[EG 2,
280	6	$\mathcal{Q}^m_\infty\leq\mathcal{Q}^m_1$
288	1	that, for each positive integer j,
288	2	$S_j=\{\qquad\qquad\qquad\qquad>\lambda+1/j\}$
288	3	is open, hence, for any positive integers i and j,
288	4	$T_{j,i}=\{(a,p):(a,r,p,s)\in S_j$ for

page	line	correction
288	6	greater than λ equals $\bigcup_{j=1}^{\infty} \bigcap_{i=1}^{\infty} T_{j,i}$.
307	7	delete this line
307	26	and $\mathcal{H}^1[p_u(E)] = 3/\sqrt{5}$ whenever $u \in U$; moreover $\mathcal{H}^1 p_u[U\Psi(I\times I)] = 2(u_1/4 + u_2)$ if $u \in S^1$ and $0 \le u_2 \le u_1/2$, $\mathcal{H}^1 p_u[U\Psi(I\times I)] = 3u_1/2$ if $u \in S^1$ and $u_1/2 \le u_2 \le u_1$, hence $\mathcal{H}^1[p_u(E)] \le 3/\sqrt{5}$ for all $u \in S^1$; since the families $\{T \cap E : T \in G_j\}$
318	26	[WH 2] or
326	3	$\Lambda^1 \mathbf{R}^n$
327	24	$S_i \cap \cup\{S_j : 0 \le j < i\} = \emptyset$, $V_i \cap \mathrm{Clos}\, S_i \subset S_i \cup G_i$ for some analytic subvariety G_i of V_i with $\dim \gamma_0(G_i) < m$ and $G_i \subset \cup\{S_j : 0 \le j < i\}$.
345	36	$\mathfrak{D}(U,Y)$
346	1	Now suppose
348	26	$\mathfrak{D}(U,Y)$
356	16	$\mathcal{H}^n$
357	17	$\mathbf{R}^n$
363	2	A (roman)
364	11	$\mathcal{D}^j(\mathbf{R}^n)$
373	29	$\mathbf{R}^n$

page	line	correction
374	16	$Q+\partial S$
383	16	Φ (lower case letter)
384	19	$e_m, Df(x)$
386	2	$<2^{-j}$ when $j>1$,
386	3	$B' = \bigcup_{j=1}^{\infty} B_j$
386	5	replace the first " B " by " B' "
386	5	between lines 5 and 6 insert the following new line: Then (4) holds with $B=B' \cap \{y:\eta(y)\neq 0\}$.
392	14	$\mathbf{R}^m$
393	25	$\mathbf{R}^m$
396	16	$\mathbf{F}_{m-1,K}(\mathbf{R}^n)$
401	12	$\mathbf{T}$ (boldface)
416	2	$\gamma = 2n^{2m+2}$
416	15	$\mathbf{B}(b,r)$
423	18	$\pm \mathbf{E}^n \llcorner A$
426	12	$\mathbf{I}^{\vee}_{m,K}(U) \neq \{(T)^{\vee} : T \in \mathbf{I}_{m,K}(U)\}$
426	16	Replace " $\mathbf{I}^2_2(\mathbf{R}^6)$ " by " $\mathbf{I}^2_{2,\mathrm{spt}\ S}(\mathbf{R}^6)$ " twice.
431	28	$\mathcal{J}$ (script)
434	9	components of $S_i \sim \cup\{G_j : i<j\leq\mu\}$.
444	20	$[(\mathscr{H}^m$
455	30	U (italic)
457	18	4.3.17 Theorem. If
462	1	when S
462	23	$(\Lambda_{k+\ell-n} g)\Xi =$
465	18	$A \cap B' = B$, $A' \cap \mathrm{Clos}(A' \sim B') \subset A$, $A' \cap \mathrm{Clos}(A' \sim A) \subset B'$,
478	26	μ (boldface)

page line correction

479 1 replace lines 1 to 7 by the following new lines:

we suppose $0<s<\infty$, σ as in 4.2.5 with $R=\mathbf{U}(0,s)$, and $\varepsilon>0$ with

$$\varepsilon+(\sigma\varepsilon)^{n/(n-1)}<\mathbf{\Gamma}^n[\mathbf{U}(0,s)\sim P]=\mathbf{\Gamma}^n[\mathbf{U}(0,s)\cap P].$$

Whenever r is sufficiently large there exist

$$X\in\mathcal{R}_{n-1}(R^n),\ Y\in\mathcal{R}_n(R^n) \text{ with } \mathbf{M}(X)+\mathbf{M}(Y)<\varepsilon,$$

$$\operatorname{spt}\left(\partial[\mathbf{E}^n L(\mu_r\circ\tau_{-b})A]-C-X-\partial Y\right)\subset R^n\sim\mathbf{U}(0,s),$$

and 4.5.2(1) yields an integer c such that

$$\left\|\mathbf{E}^n L(\mu_r\circ\tau_{-b})A-\Theta^{n-1}(\|T\|,b)\mathbf{E}^n LP-Y-c\mathbf{E}^n\right\|\mathbf{U}(0,s)\leq[\sigma\mathbf{M}X\mathbf{M}\mathbf{U}(0,s)]^{n/(n-1)};$$

letting φ_r and ψ_r be the characteristic functions of $(\mu_r\circ\tau_{-b})A$ and P we obtain

$$\int_{\mathbf{U}(0,s)}|\varphi_r-\Theta^{n-1}(\|T\|,b)\psi-c|\,d\mathbf{\Gamma}^n<\varepsilon+(\sigma\varepsilon)^{n/(n-1)},$$

$$\varphi_r(x)-c=0 \text{ for some } x\in\mathbf{U}(0,s)\sim P,$$

$$\varphi_r(x)-\Theta^{n-1}(\|T\|,b)-c=0 \text{ for some } x\in\mathbf{U}(0,s)\cap P,$$

and infer $c\in\{0,1\}$, $\Theta^{n-1}(\|T\|,b)+c\in\{0,1\}$, hence $\Theta^{n-1}(\|T\|,b)=1$, $c=0$. Thus u is an exterior normal of A at b. We conclude

483 5 $\bar{\mathbf{R}}$

page line correction

499 Insert at the bottom of this page the following:

and for each $\varepsilon>0$ there exists $\delta>0$ such that
$\rho^{-n}\|\partial(T-\mathbf{E}^n\llcorner L)\|\mathbf{U}(b,\rho)\leq\varepsilon$ whenever $0<\rho\leq\delta$,

$$\left(\rho^{-n}\int_{\mathbf{U}(b,\rho)}|f-L-t_\rho|\,d\mathcal{L}^n\right)^{1/\beta}\leq\sigma\varepsilon\rho \text{ whenever } 0<\rho\leq\delta,$$

$$\alpha(n)^{1/\beta}|t_r-t_s| = \left(s^{-n}\int_{\mathbf{U}(b,s)}|t_r-t_s|^\beta d\mathcal{L}^n\right)^{1/\beta}$$
$$\leq \sigma\varepsilon s + (r/s)^{n/\beta}\sigma\varepsilon r \leq (1+2^{n-1})\sigma\varepsilon r$$

whenever $0<r/2\leq s\leq r\leq\delta$, hence

$$|t_\rho-f(b)-L(b)| \leq \sum_{i=0}^{\infty}|t_{2^{-i}\rho}-t_{2^{-i-1}\rho}| \leq \sum_{i=0}^{\infty}c\varepsilon 2^{-i}\rho = 2c\varepsilon\rho$$

whenever $0<\rho\leq\delta$, with $c=\alpha(n)^{-1/\beta}(1+2^{n-1})\sigma$,

502 3 $\mathcal{H}^{n-1}(E)$

503 12 all w are absolutely

510 14 are the usual

511 24 $[\log(e-1)]^{1-n}/(n-1)$

515 23 In 5.2.19 we

517 10 $\Phi_a(\alpha)=\Phi(a,\alpha)$

521 33 $\mathbf{H}_m$

522 27 $T-Q\in\chi$

525 23 neighborhood of $B\cap\operatorname{spt}\partial T$ and

526 17 $\Xi_z = \frac{1}{2}\rho_z^*\circ D^2h_0(z)\circ\rho_z$

526 20 $+\frac{1}{2}t^2D^2h_0(z)\circ\rho_z$

532 7 $f_\ell = Y_\ell\circ f$

533 15 $\leq\gamma^2$

539 20 $|D^{p+j}f|$

page	line	correction
539	23	$\|D^{i+q}f\|_{b,\rho/2}$
541	21	converge to a Υ harmonic
544	5	$\mathbf{e}_i$ (boldface)
547	9	is strongly elliptic
560	22	sup im Φ (capital)
568	13	$\kappa^{2/m}$
569	3	$\kappa^{2/m}$
569	19	$\kappa^{2/m}$
573	28	replace " 96 " by " 192 "
574	2	replace " 96 " by " 192 "
574	10	replace " 3 " by " 4 "
574	18	conjunction with (2) and (3).
575	5	$\mathbf{p}$ (boldface)
575	21	$6E+\kappa\Omega+\kappa\omega$
576	15	replace " D_j " by " D_i "
580	2	$\|\Phi($
582	19	replace " 5 " by " 6 "
582	27	replace " 5 " by " 6 "
583	16	replace " 5 " by " 6 "
585	3	replace " 5 " by " 6 "
589	26	S^2_ν S^1_ν S^2_ν (insert subscripts)
591	7	$(S^q_\nu\llcorner$
592	4	$\mathbf{p}$ (boldface)
592	21	$h_{\nu,1}$
592	30	replace " M_2 " by " M_4 "
593	28	$h_{\nu,1}$
594	27	$G_\nu\#$

page	line	correction
595	18	$E+\kappa^2\sigma tr$
595	24	$\alpha\kappa^2 E$
596	13	$\alpha\kappa^2 E$
596	16	$t^{8m}\alpha\kappa E+2^{m+2}\sigma tr[\kappa\alpha(m)+t^{8m}\alpha\kappa^2 E]$
597	10	$\alpha(m)+\kappa^2\alpha(m)/5$
597	23	replace " 4^4 " by " 4^3 "
598	14	$D\tilde{\Phi}_a(p^*)$
599	15	$=-\tilde{\Phi}$
599	23	replace " $32\mu^6$ " by " $2^4\mu^4$ "
601	18	$M^{90m^2}\kappa^3$
601	19	$M^{205m^2}\kappa^4$
601	22	$(\Gamma_5+8\Gamma_4)\kappa^2+$
601	24	κ^3
601	25	$\lambda=\Gamma_{18}\Gamma_{19}M(4\mu^6)^{5m}=\Gamma_{18}\Gamma_{19}\,2^{10m}\mu^{30m}M,$
601	27	replace " κ^2 " by " $\kappa^3 M^{1+m}$ "
601	28	replace " κ^2 " by " $\kappa^3 M^{m+4}$ "
601	31	replace " κ^3 " by " $\kappa^4 M^{m+4}$ "
601	32	replace " κ^2 " by " κ^3 " (twice)
602	23	Then we apply
602	24	replace " 4^4 " by " 4^3 "
602	29	replace " 2^9 " by " 2^7 "
603	5	$\leq\Gamma_{17}(2\mu^2)^{8m}\kappa(d_2+\kappa^2 4^3\mu^8 2\mu^2 r_2)+(8\mu^2)^m(4\mu^2)^m d_2$
604	24	$M^{90m^2}\kappa^3$
605	10	$+\kappa^2\sigma_4$
605	11	$+\kappa^2 2^5$
605	21	$M^{205m^2}\kappa^4$

page	line	correction
606	24	$\Delta_4=\Gamma_{20}M^{90m^2}\kappa^3$, $\Delta_5=2\Gamma_{21}M^{205m^2}\kappa^4$,
607	16	$G_\nu = \mathbf{1} - \mathbf{q}^*\circ g_\nu\circ\mathbf{p}$
608	25	$c_\nu - g_\nu(b_\nu)$
610	20	$\tau_{(b,c)}{}^{\#}\Phi$
616	19	is contained
618	11	replace " S_j " by " S_i "
619	17	$spt^2S\sim spt^2\partial S$
619	18	spt^2S
619	26	of 5.3.20 becomes
619	29	$spt^3\partial T$
623	14	$\|\vec{T}\|LDu\|$ (insert " | ")
630	23	$\lim_{i\to\infty}$
633	7	$\mathbf{1}$ (boldface)
644	23	$\mathbf{R}^n$
652	12	$\langle\vec{T}(z),A^\mu\rangle$
654	8	$q \geqslant 3$
654	13	<u>class</u> q-1
664	43	(1955)